LECTURE 02

Terminologies in Theory of Computation

Course Code : CSE 3103
Course Title : Theory of Computation

Md. Sozib Hossain
Lecturer, CSE, RUET
sozib.hossain@cse.ruet.ac.bd

Modeling a Computer

<u>Computer/Automaton</u>

Computing Device or Machine. For example: Laptop, Desktop, Calculator etc.

Since, we can say typical calculator is a computer.

So,

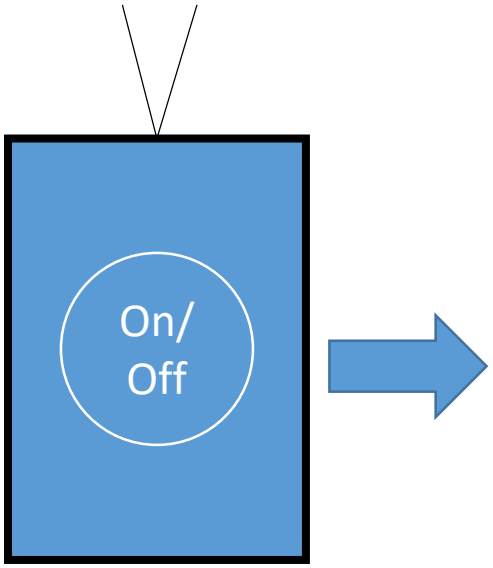
- **Is calculator that perform only addition a computer?**

  -Yes.

- **Is calculator that has only power button a computer?**

  -Yes.

Modeling a Computer (Cont…)



States = {On, Off}
Input = {Press, Not Press}
Transition Function =

	Input	
States	Press	Not Press
On	Off	On
Off	On	Off

Initial State = {Off}

Fig 01: Simple computer with one button.

Physical Computer

Mathematical Model of Computer

By using DFA

COMPUTATIONAL MODEL

<u>There are several mathematical models of computation that are used to design theoritical computers such as:</u>

- Finite Automata
  - Deterministic Finite Automata (DFA)
  - Non-deterministic Finite Automata (NFA)
- Push Down Automata (PDA)
- Context Free Grammar (CFG)
- Turing Machine (TM)

BASIC TERMINOLOGY

SET

A set is a group of objects represented as a unit.

$S = \{5, 7, 8\}$ Here, 5 7 and 8 are called **elements or members**.

- **A** is a **subset** of **B**, written $A \subseteq B$ and A is a **proper subset** of B, written $A \subset B$, if A is a subset of B and not equal to B.
- The **order** of a set doesn't matter, nor does **repetition** of its members. $\{1, 2, 3\}$ & $\{2, 3, 1\}$ - equivalent | $\{1, 2, 3\}$ & $\{1, 2, 2, 3, 3\}$ - equivalent.
- An infinite set use "**...**" notation to mean "continue the sequence forever".
- The set with zero members **-empty set**, written as $\emptyset$. A set with one member **-singleton set**, and a set with two members **-unordered pair**.
- Two sets **A** and **B**, the **union** written as $A \cup B$, the **intersection** written as $A \cap B$, the **complement** of A, written as $\bar{A}$.

BASIC TERMINOLOGY (CONT…)

SEQUENCES AND TUPLES

A sequence of objects is a list of these objects in some order.

$$(7, 21, 57)$$

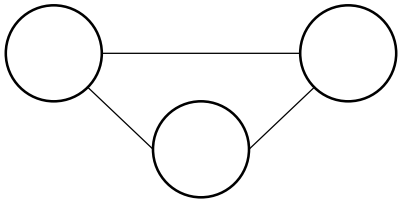
- **Order & Repetition** matters in a sequence.
- As with sets, sequences may be finite or infinite. Finite sequences often are called **tuples.** A 2-tuple is also called an **ordered pair.**
- The **power set** of A is the set of all subsets of A. If **A** is the set $\{0, 1\}$, the power set of A is the set $\{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$.
- The **Cartesian product or cross product** of A and B, written $A \times B$, is the set of all ordered pairs wherein the first element is a member of A and the second element is a member of B. If $A = \{1, 2\}$ and $B = \{x, y, z\}$,

$$A \times B = \{ (1,x), (1,y), (1,z), (2,x), (2,y), (2,z) \}.$$

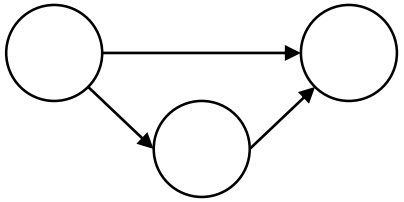
BASIC TERMINOLOGY (CONT…)

GRAPHS

An undirected graph, or simply a graph, is a set of points with lines connecting some of the points. The points are called **nodes or vertices**, and the lines are called **edges**.



- We may allow an edge from a node to itself, called a **self-loop**.
- Label the nodes and/or edges of a graph, which then is called a **labeled graph**. Nodes are labeled by **state name**. Edges are labeled by input **symbol**.
- A **directed graph** has arrows instead of lines.



BASIC TERMINOLOGY (CONT…)

SYMBOL & ALPHABET

Symbol is user defined entity or it is collection of letters, digits, pictures and special characters.

{a-z, A-Z, 0-9, *, #, @…}

An alphabet is a finite set of symbols denoted by Σ.

For Binary number

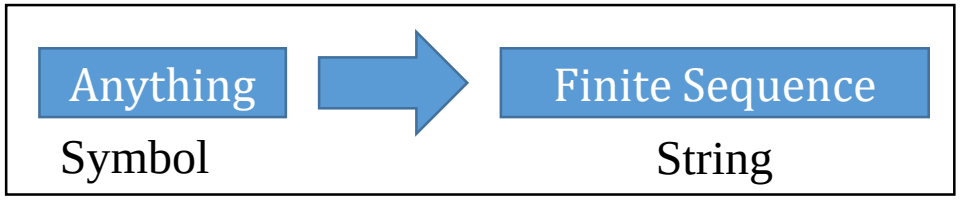
$$\Sigma = \{0,1\}$$

For Hex Number

$$\Sigma = \{0\text{-}9, A\text{-}F\}$$

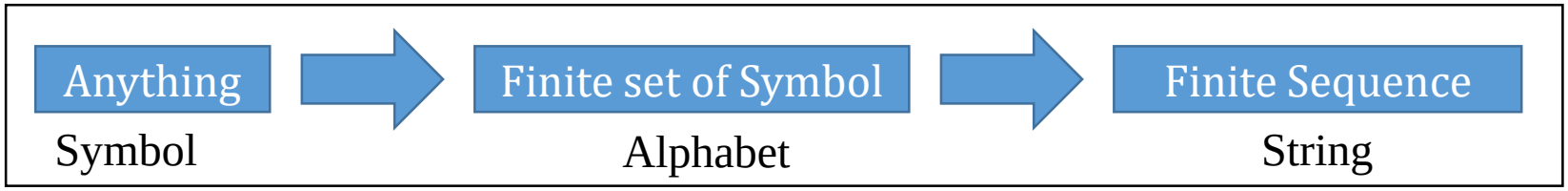
BASIC TERMINOLOGY (CONT…)

STRING



In general

In theory of computation



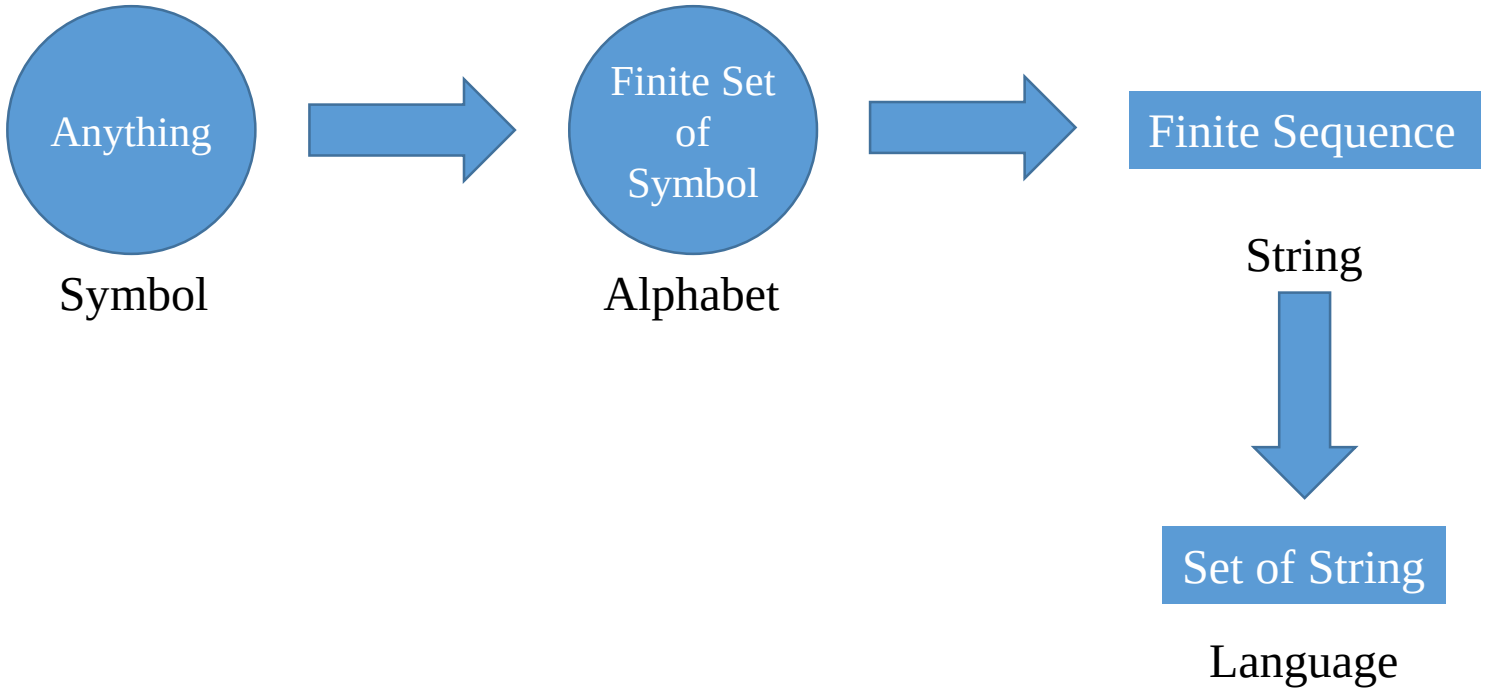
A string is a finite sequence of symbols selected from an alphabet. String is denoted by **S** and the length of a string is denoted by **|S|.**

For example, if an alphabet is {a, b, c, d} then one string from this alphabet will be **S** = bacd and the length of this string is **|S|** = 4.

BASIC TERMINOLOGY (CONT…)

LANGUAGE

A language is a set of string that satisfy specific criteria or solve a specific problem.



BASIC TERMINOLOGY (CONT…)

LANGUAGE

For Example, all three digits binary numbers.

Symbol: {a-z, A-Z, 0-9, *, #, @...}

Alphabets: {0, 1}

String: {0, 1, 00, 01, 10, 10, 11, 000, 001, 010, 011, . . .}

Language: {000, 001, 010, 011, 100, 101, 110, 111}

BASIC TERMINOLOGY (CONT…)

MORE LANGUAGE EXAMPLE

1. All binary even numbers.
2. All three digit binary even numbers.
3. All binary number starting with 101.
4. All binary number ending with 111.
5. All binary number starting with 111 and ending with 101.
6. All binary numbers that contains even number of 0's.
7. All binary numbers that contains odd number of 1's.
8. All binary number that contain even number of 0's and even number of 1's.